One device, two complementary ways of thinking

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WEBSTER ARCHITECTURE physical components	COURSE ANALYTICAL SPINE questions we keep asking
measurand → sensor/transducer → conditioning → output/display (+ optional auxiliary elements)	measurand → transduction → conditioning → integration → performance → safety

measurand	→	**measurand**
sensor / transducer	→	**transduction**
signal conditioning	→	**conditioning**
output + auxiliary	→	**integration**

Performance and safety are explicit course questions — not extra physical boxes.

Take-away: architecture tells us what is there; the spine tells us what to ask next.

Webster anchor: Ch. 1 §1.2, pp. 3–5. Medical constraints: §1.4, pp. 7–10.

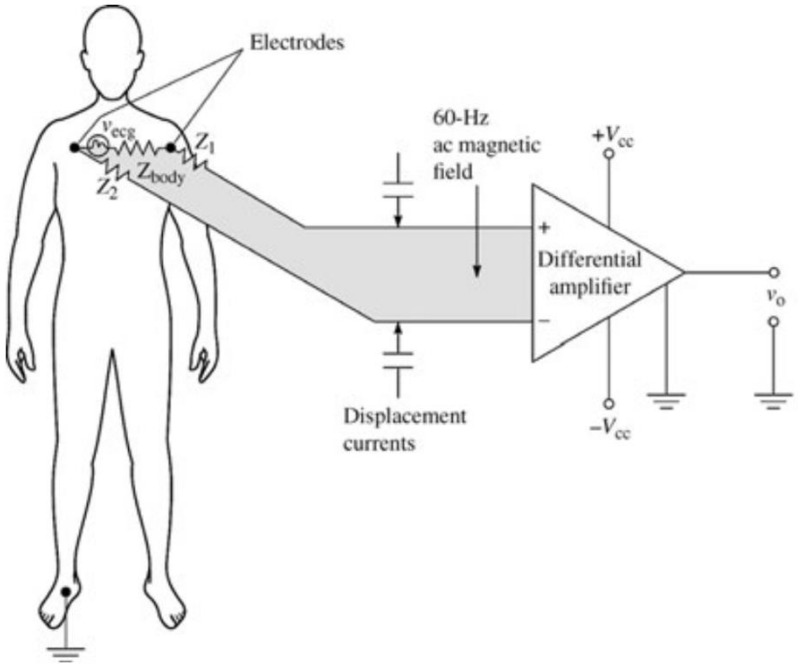
Every medical device lives under constraints

SAFETY	BIOCOMPATIBILITY	OPERATING ENVIRONMENT	REGULATION
Do not harm the patient or operator. Energy, current, heat, and failure modes matter.	Anything contacting tissue must behave acceptably in that biological environment.	Motion, moisture, temperature, users, infrastructure, and maintenance change the design.	A medical device must meet the requirements that govern its intended use and risk.

Teaching question: for a wearable ECG patch, which constraint is most likely to bind first?

Webster 5e: Ch. 1 §1.4 “Medical Measurement Constraints,” pp. 7–10.

Wanted signal vs. everything else



DESIRED INPUT

ECG voltage from the patient → **ECG recording system** → recorded ECG trace

INTERFERING INPUT

50/60 Hz mains field induces unwanted voltage

MODIFYING INPUT

Cable orientation / electrode–skin impedance changes mains coupling

How we compensate for these effects → §1.7 and Week 5.

Webster 5e: Ch. 1 §1.6 “Interfering and Modifying Inputs,” pp. 10–12; Fig. 1.2, p. 11. §1.7 begins p. 12.

Put the pulse oximeter on the chain

MEASURAND	TRANSDUCTION	CONDITIONING	OUTPUT / DISPLAY
Arterial oxygen saturation (SpO_2) inferred at the fingertip	Red + infrared light through tissue photodetector → current	Amplify · filter · isolate pulsatile component digitize	Compute SpO_2 from two-wavelength ratio show number + waveform

Performance limits to notice now: motion artifact · low perfusion

Week 1 question: where in the chain would each problem first become visible?

Webster 5e: Ch. 10 §10.6, pp. 610–613; Fig. 10.19, p. 612. Motion / low-perfusion examples are lecturer-applied performance examples.

3.1. Datasheet handout — clinical pressure transducer

Spec	Value
Range	−60 to +200 mmHg
Static accuracy	±2% of reading
Zero drift	≤ 1 mmHg/hr
Frequency response	flat (±5%) to 20 Hz
Natural frequency (f_n)	≈ 25 Hz
Damping ratio ζ	≈ 0.3
Requirement	**flat to 30 Hz, no visible systolic overshoot**

Task: using this week's static and dynamic characteristics, judge whether this device is fit for the stated clinical purpose. Justify your judgement in one or two sentences.

Source: lecturer-constructed illustrative specification sheet — not manufacturer data. Range anchor: Webster ch. 1, p.23

3.2. Datasheet handout — clinical thermistor probe

Spec	Value
Range	25–45 °C
Static accuracy	±0.1 °C
Resolution	0.01 °C
Time constant τ	2.0 s
Requirement	**settle within 5% of a step change within 10 s**

Task: using this week's static and dynamic characteristics, judge whether this device is fit for the stated clinical purpose. Justify your judgement in one or two sentences.

Source: lecturer-constructed illustrative specification sheet — not manufacturer data

1.2. Strain gauge and gauge factor

Stretching a wire makes it longer and thinner. Its resistance rises.

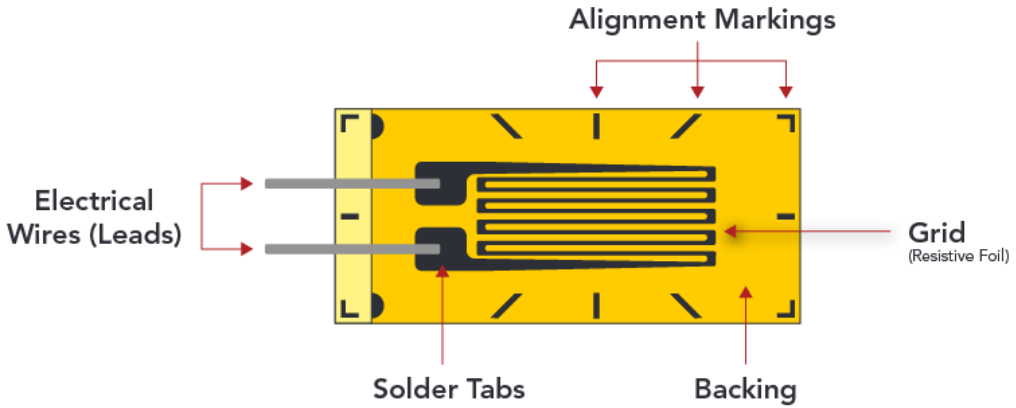
Gauge factor: G = (ΔR/R) / (ΔL/L)

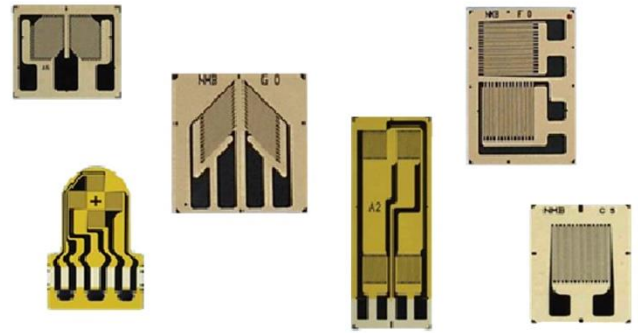
G = how much resistance changes per unit stretch; ΔL/L = the stretch (strain).

Strength: Force / pressure → small fractional ΔR
Small signal → bridge read-out
Limitation: temperature sensitivity

Real devices: the strain gauge on a blood-pressure sensor, and the strain-gauge chest strap that measures breathing.

Strain-gauge geometry and deformation



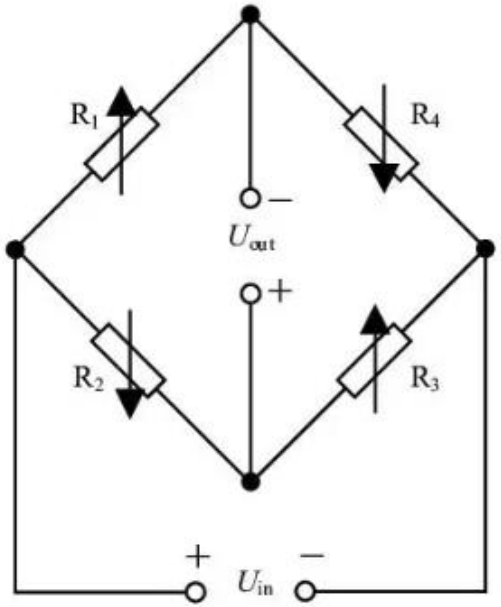


Source: Webster §2.2, pp.96–101

1.5. The Wheatstone bridge & a.c. bridge – turning a small change into voltage

Two voltage dividers: balance → $U_{out} = 0$.
A sensor unbalances an arm; the difference is the signal.

High sensitivity with amplification effect



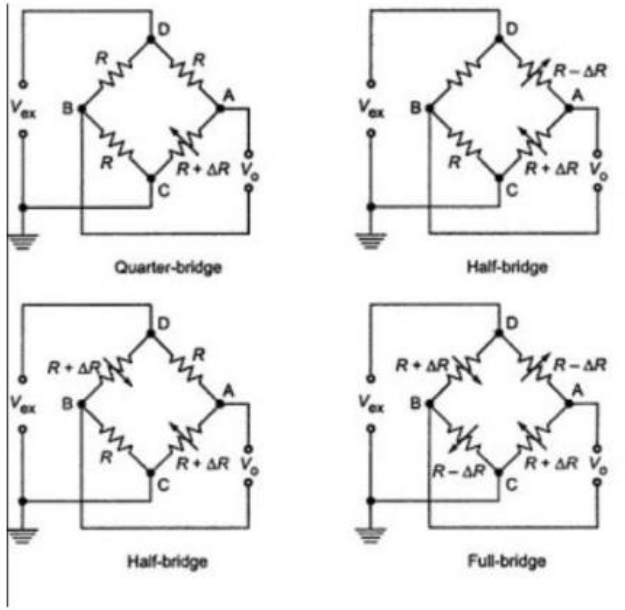
Wheatstone-bridge circuit

https://youtube.com/shorts/ImOjtWkTjBk?si=6G8-gxy0fHmvgiK9

At Balance: $\frac{R_1}{R_2} = \frac{R_4}{R_3} \rightarrow U_{out}/v_0 = 0$

- We replace one or more resistors with resistive sensors in arms (e.g., strain gauges)

Quarter / half / full: 1 / 2 / 4 active arms



Four active arms:
$\Delta v_o = (\Delta R/R_0) \cdot v_i$

R_0 = initial resistance

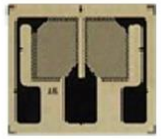
ΔR = change caused

https://www.youtube.com/watch?v=-gcBVaCIKhE

Source: Webster §2.3, pp.102–103

1.5. Worked example: bridge read-out on strain gugage

$R_o = 350\ \Omega$ $v_i = 5.0\ V$ $G = 2.0$ $\varepsilon = \Delta L/L = 1.0 \times 10^{-3}$



$G = (\Delta R/R) / (\Delta L/L)$

$$\Delta R = G\varepsilon R_o = 2.0 \times 10^{-3} \times 350 = 0.70\ \Omega$$

Configuration	Active arms	Output	
Quarter bridge	1	2.50 mV	$\Delta v_o = 1/4(\Delta R/R_o)\cdot v_i$
Half bridge	2	5.00 mV	$\Delta v_o = 1/2(\Delta R/R_o)\cdot v_i$
Full bridge	4	10.0 mV	$\Delta v_o = (\Delta R/R_o)\cdot v_i$

This is a common read-out for strain-gauge pressure sensors.
More active arms give a larger signal and better cancel temperature —
but only if the gauges are matched and arranged so heat effects cancel.

Source: Webster §2.3 (Eq. 2.6), p.102

2.1. Static Load – A held force does not give a held reading

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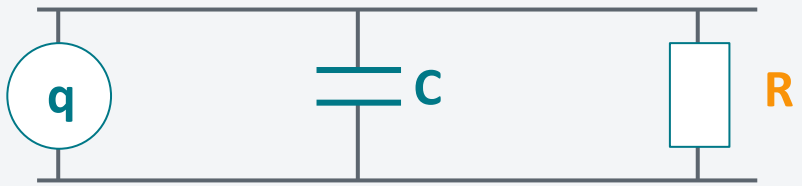
PRESS → **HOLD** → **READING DECAYS**

What to observe

Press → transient; hold → decay

Press gives a voltage transient; hold and it decays through the **leakage low-corner**. Therefore, a piezo cannot report a static load.

Equivalent circuit



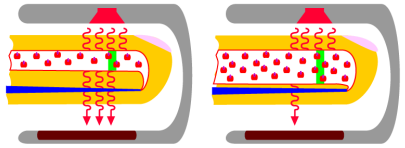
A piezoelectric sensor reports changing load, not a static load.

Source: Webster §2.7, Fig. 2.12;

2.3. Optical transduction

Shine light through or onto tissue. Measure the light that reaches the detector.









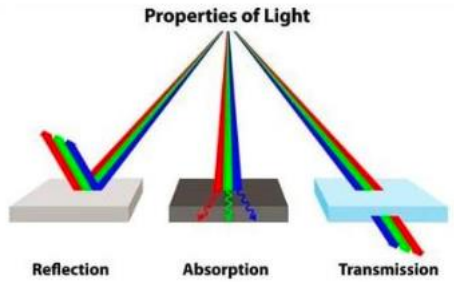
Light source	→	Tissue / optical path	→	Filter	→	**Photodiode / phototransistor**

Strength

The sensor does not touch the patient electrically.
Example: the **pulse oximeter** clipped on fingertip.

Dominant limitations

Ambient light, motion and calibration can change the measured signal.



- **Reflectance Oximetry** (Light Reflection) – e.g., smart watch
- **Transmittance Oximetry** (Light Absorption) – e.g., oximeter

Pulse oximetry uses this pattern. Device-level treatment returns in Week 7.

Source: Webster §2.14–2.19, pp.133–150

2.5. Comparison: properties that drive selection

Family	Response	Strength	Dominant limit	Biomedical example
Strain gauge	Static + dynamic	Force / pressure route	Temperature; small signal	BP diaphragm; chest strap
LVDT	Static + dynamic	Frictionless; high resolution	a.c. drive; magnetic materials	Pressure core; displacement
Capacitive	Static + dynamic	Non-contact; tiny displacement	Stray capacitance; cable effects	Pressure; touch / proximity
Piezo-electric	Dynamic only	Fast; self-generating	No dc; charge read-out	Heart sounds; ultrasound
Thermistor	Static + dynamic; τ set by thermal mass	Sensitive; low cost	Nonlinear	Temperature measurement
Thermo-couple	Static + dynamic; τ set by thermal mass	Wide temperature range	Small signal; reference junction	Temperature measurement
Optical	Static + dynamic	No electrical patient connection	Ambient light; motion; calibration	Pulse oximetry

Source: BME 6111 Week 3 lecturer synthesis (authored comparison)

3.1. Knee-angle selection activity

Wearable rehabilitation monitor: knee joint angle during gait

0–140° range ≤1° resolution ~10 Hz signal bandwidth
Worn on the leg: motion, sweat and limited power

Candidate	Your reasoning
Resistive flex sensor	Fit, limitation and evidence
Potentiometer at the joint axis	Fit, limitation and evidence
Inductive angle sensor	Fit, limitation and evidence
Capacitive angle sensor	Fit, limitation and evidence

Suggestion: Use the internet/AI to find out more about what they are **BUT NOT** in the exact context

Source: BME 6111 Week 3 lecturer-authored activity

3.1. Debrief: two defensible starting choices

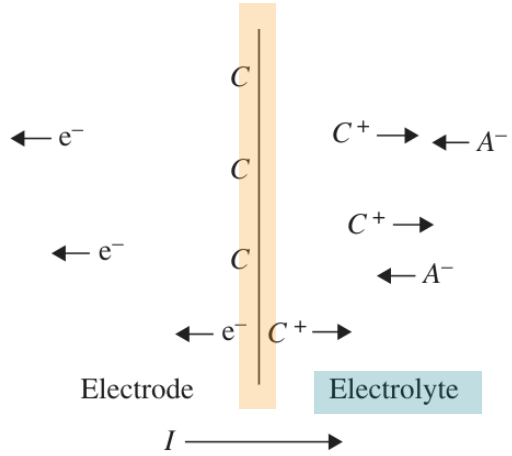
Candidate	0–140° range	≤1° resolution	~10 Hz bandwidth	Motion / sweat / power
Resistive flex sensor	**Potentially good wearable range**, provided bend range covers the required motion	Must verify whether a 1° change produces a repeatable enough resistance change; hysteresis can matter	Likely plausible for gait-scale motion, but actual response must be verified	Very wearable and low-profile; however attachment, repeated bending, sweat and mechanical hysteresis can shift the reading
Potentiometer at joint axis	**Potentially strong fit** if its mechanical rotation covers ≥140°	**Potentially strong** because angle maps directly to wiper position; actual resolution must be verified	Usually plausible for slow mechanical joint motion; verify device response	Low-complexity resistive read-out and modest power; but rigid alignment at the joint and moving contact can be awkward in a wearable
Inductive angle sensor	Can measure displacement/angle over a designed range	Can offer high resolution	Dynamic response may be adequate, but must be checked	Needs a.c. excitation/read-out and may be affected by nearby magnetic material; less attractive under a limited-power wearable constraint
Capacitive angle sensor	Geometry can be designed for angular displacement	Can be highly sensitive to small changes	Usually capable of fast response	Weakest point here: stray/parasitic capacitance and changing geometry/dielectric conditions from sweat, clothing and body movement; also requires electrical interrogation

Source: BME 6111 Week 3 lecturer synthesis (resolved judgement)

2.1 How charge transfers between tissue and metal

Current is carried by electrons in the metal and by ions in the electrolyte.

Electrolyte	a liquid or gel containing mobile ions
Interface	the boundary where metal meets electrolyte
Oxidation	a reaction that releases electrons
Reduction	a reaction that accepts electrons



C = the book's symbol for a **metal atom** (not carbon)
I = conventional current

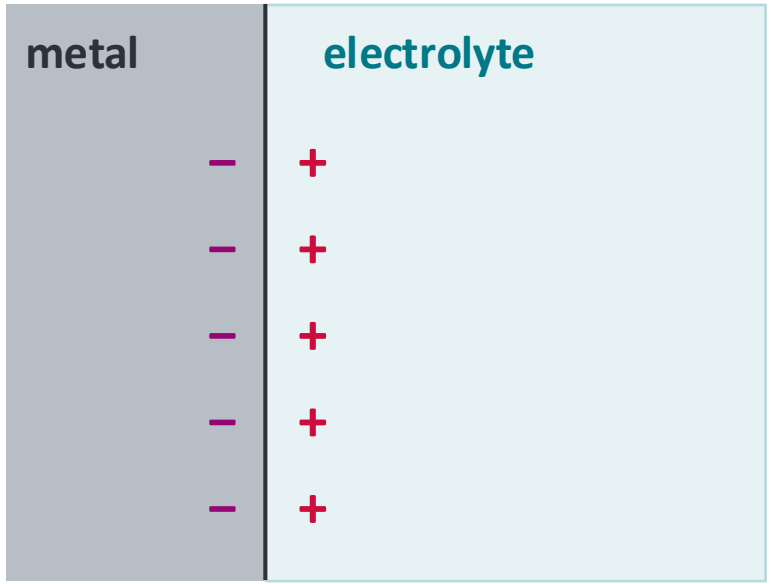
1. **Electrolyte:** cations (C^+) and anions (A^-) carry current
2. **Boundary:** reactions pass charge across
3. **Metal side:** electrons (e^-) carry current

Electrode reactions connect ionic conduction in the body to electronic conduction in the lead.

Source: Webster §5.1, Fig 5.1, p. 268

2.2 An electrode develops its own interface voltage

Charge separation at the boundary creates a voltage, even with no net current.



(which side is + depends on the metal)

1. Reactions redistribute charge near the boundary.
2. Opposite charges collect on the two sides.
3. The equilibrium voltage is the **half-cell potential, E_{hc}**.

Set by: electrode material · ion concentration · temperature

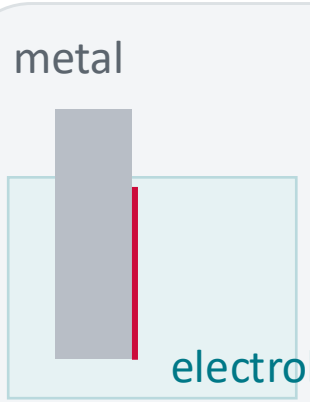
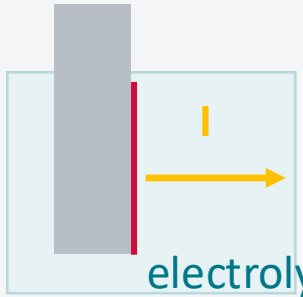
E_{hc}	electrode interface
v_m	cell membrane (Block 1)

An electrode adds its own voltage on top of the biological voltage we want to measure.

Source: Webster §5.1–5.2, pp. 268–271

2.2 Polarization is a change in electrode voltage

Current through the electrode can pull its interface voltage away from E_{hc}.

NO NET CURRENT	CURRENT FLOWS
 Interface voltage = E_{hc} (equilibrium)	 Interface voltage = E_{hc} + overpotential

Polarization = this departure from equilibrium · **Overpotential** = the size of the voltage departure

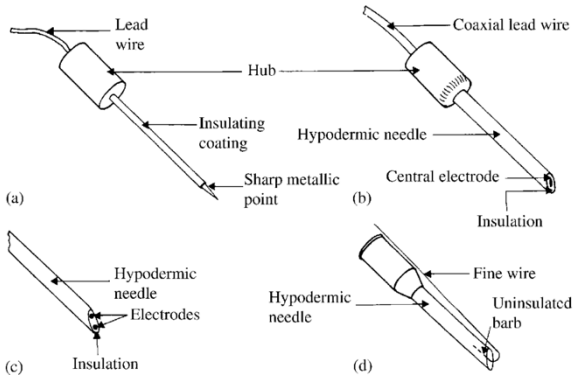
A more polarizable interface changes its voltage more readily when current passes through it.

Source: Webster §5.1–5.2, pp. 268–271

2.5 Recording location determines what an electrode measures

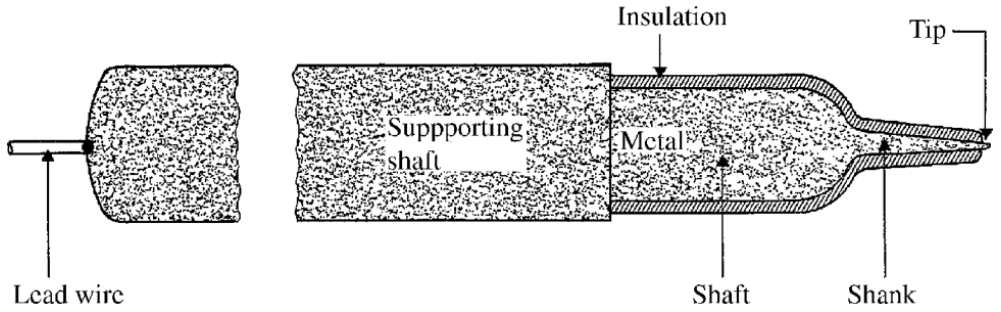
Under the skin: local activity. Inside a cell: that cell's membrane voltage.

INTERNAL / NEEDLE — under the skin



Contact lies beneath the skin
Records local muscle activity
Recording contact = the exposed tip or wire

MICROELECTRODE — intracellular example



A fine exposed tip makes a very local contact
Insulation covers the shaft
Tip inside the cell, against an outside reference

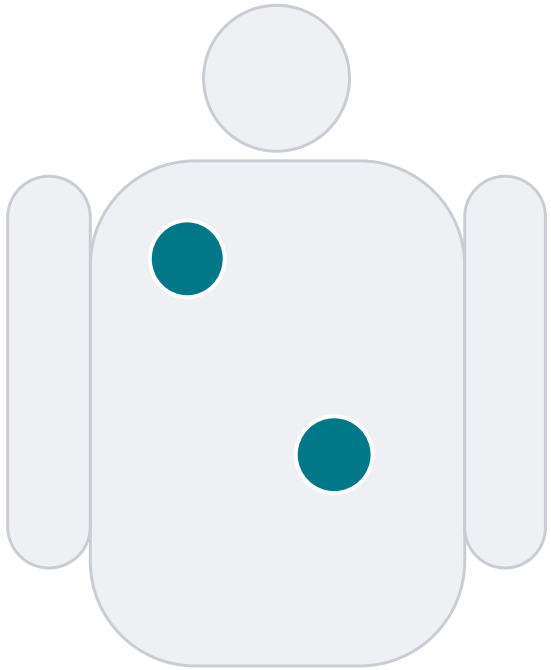
Location decides what you see: a surface signal, local tissue activity, or one cell's membrane voltage.

Source: Webster §5.7, Fig 5.13, p. 305; §5.9, Fig 5.17, p. 312

3.1 Design brief: stable ECG during movement for several days

Each requirement creates a different demand on the electrode and its skin contact.

Requirement	What it means for the design
ECG amplitude: 0.5–4 mV	Electrode voltage changes can be as large as the ECG itself
Monitoring band: about 0.67–40 Hz	Slow baseline changes can overlap the useful ECG band
Walking and exercise	The contact must stay mechanically stable during movement
Several days of wear	Gel dry-out, adhesion and skin tolerance become important



Chest-worn ECG patch: two contacts

The electrode must keep a stable contact during movement without failing after several days.

Source: BME 6111 Week 4 lecturer-authored activity

3.1 The candidate electrodes

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Every candidate has a reason to consider it and a main concern.

Candidate	Reason to consider	Main concern	Photo
Pregelled foam Ag/AgCl (liquid/wet gel)	Low, stable impedance; established option	Gel can dry and irritate skin in long wear	foam disc
Dry Ag/AgCl	No added gel to dry out; keeps the Ag/AgCl material	Contact depends on pressure, fit and skin condition	dry Ag/AgCl
Hydrogel-adhesive Ag/AgCl	Adhesive ionic contact can stay stable during movement	Hydrogel and adhesive can dehydrate or irritate skin	hydrogel patch
Dry metal / textile	Can be thin, flexible and reusable	Depends on material, pressure and movement	metal/textile
Capacitive / non-contact	No gel and no direct conductive contact	Motion changes coupling; very high-impedance inputs needed	capacitive

Capacitive: an insulating layer separates electrode and skin, so the signal couples without conductive contact.

Source: BME 6111 Week 4 lecturer-authored activity

3.2 Model comparison: what each candidate gains and loses

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All five can detect an ECG; they differ in contact stability and long wear.

Candidate	Fit to the brief	Main limitation	Expected artifact
Pregelled foam Ag/AgCl	Stable, low impedance	Gel dries; may irritate	Drift as contact changes
Dry Ag/AgCl	No gel to dry out	Needs firm fixation	Motion baseline wander
Hydrogel-adhesive Ag/AgCl	Stable adhesive contact	May dry out or irritate	Drift if adhesion fails
Dry metal / textile	Flexible, reusable	Impedance varies with motion	Contact-motion artifact
Capacitive / non-contact	No gel; insulated contact	Demanding input electronics	Motion coupling artifact

Motion-related baseline disturbance remains the main artifact; design changes its likelihood and severity.

Source: BME 6111 Week 4 lecturer synthesis (resolved judgment)

3.2 Model decision: the priority determines the choice

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Assume the priority is a stable ECG while walking, over several days.

Decision element	Model answer
Priority	Stable ECG while moving, over several days
Electrode	**Hydrogel-adhesive Ag/AgCl**
Why it fits	Ag/AgCl polarizes little; hydrogel holds contact
Remaining limitation	Hydrogel/adhesive changes; may irritate skin
Predicted artifact	Baseline wander, worse if adhesion fails
Design response	Stable placement, skin prep, fixation

Also defensible, with a different priority

Pregelled Ag/AgCl: short-term signal stability comes first.

Dry Ag/AgCl: avoiding gel dry-out comes first, with secure fixation.

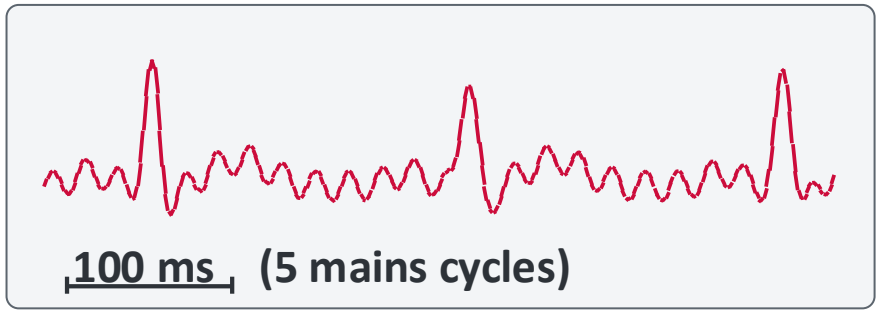
Dry metal/textile or capacitive: possible, with a clear plan for motion and input electronics.

A defensible answer links priority, choice, evidence, remaining limitation and predicted artifact.

Source: BME 6111 Week 4 lecturer synthesis (resolved judgment)

2.2 WE 2 — solved on the mains-pickup trace

The four-step method, worked on the demonstration trace.



Step	On the demonstration trace
Describe	regular ripple, 5 cycles in 100 ms → 50 Hz, constant, present between beats
Place	Family 1, periodic mains pickup
Narrow	the ripple suggests mains pickup; test electrode contact and lead routing to distinguish the possible coupling paths
Fix	re-seat electrodes for lower, balanced Z_e; shield and twist leads; a driven-right-leg loop needs an added electrode; notch only as a last resort

Verify: the 'after' trace shows the ripple gone and the ECG unchanged.

Source: lecturer-constructed worked diagnosis; after Webster ch. 6 §6.3

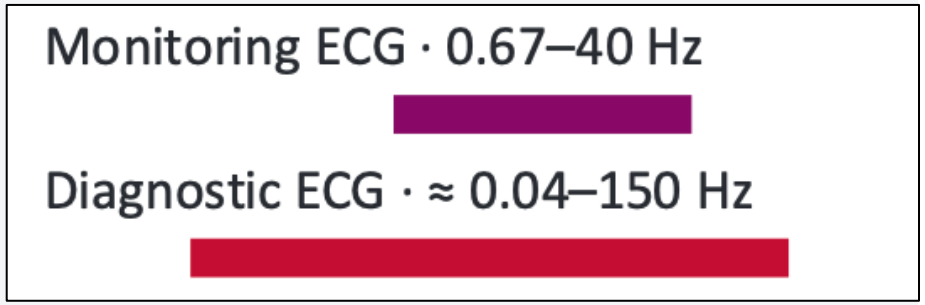
2.3 WE 3 — sampling and resolution for the patch

Sampling rate controls time detail; gain sets the ADC step size referred to the electrodes.

(a) Sampling rate $(f_s \geq 2 \times f_{max})$

- Monitor ECG: f_{max} = 40 Hz → f_s ≥ 80 Hz;
- Diagnostic f_{max} = 150 Hz → f_s ≥ 300 Hz.

At 400 samples/s, Nyquist is 200 Hz (half of sampling rate); the analog filter must attenuate content above 200 Hz.



(b) Resolution referred to the electrodes

12-bit ADC, 0–3.3 V; illustrative total gain **G_{total} ≈ 600**, chosen for headroom. (S10–S11)

ADC step q = 3.3 V / 4096 = 0.806 mV (S30); at the electrodes (before amp): q / G ≈ 1.34 µV.

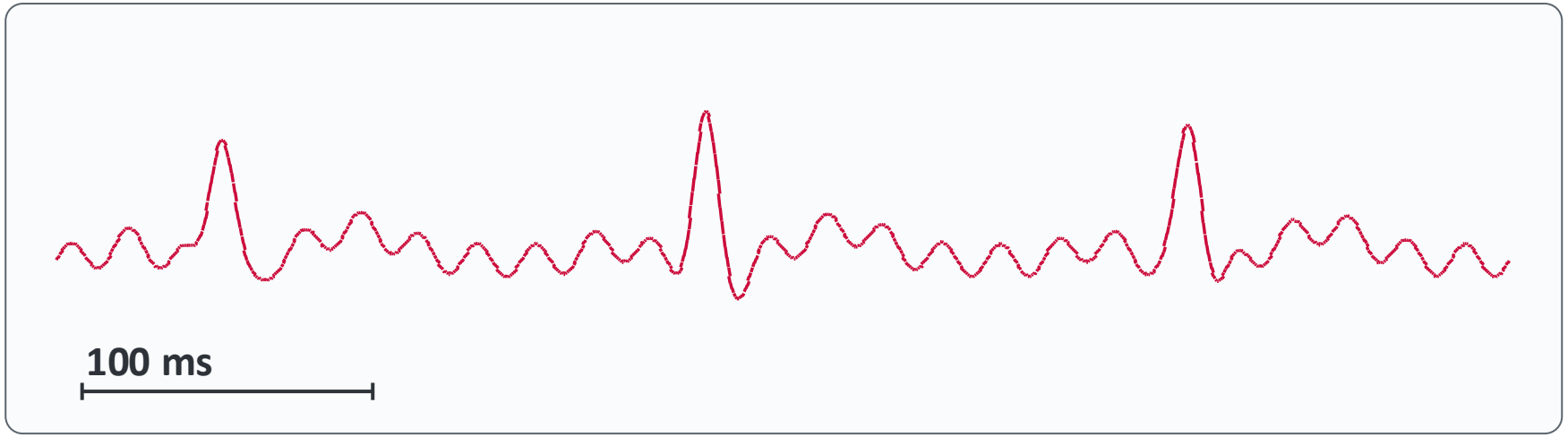
How many steps we need to record a 1 mV electrode change?

It becomes 600 mV at the ADC: 600 mV / 0.806 mV ≈ 745 steps.

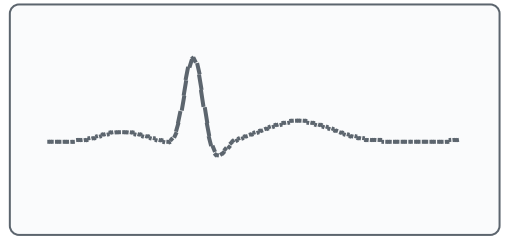
200 Hz limits frequency; 745 counts voltage steps for a 1 mV change, not time samples.

Source: lecturer-constructed; sampling rate after Webster ch. 3 Example 3.11, p. 186

3.1 Clinic — Trace T1



one recording from the patch · time scale marked below



clean reference beat

Recording circumstance

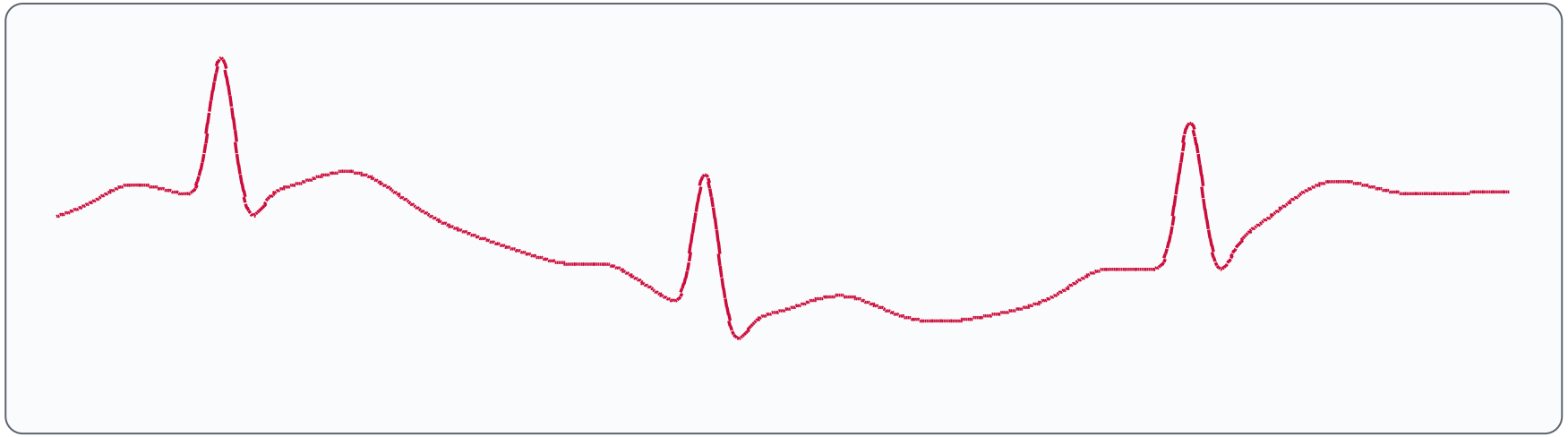
The wearer is still. The regular ripple persists between heartbeats; about five cycles occur in 100 ms.

Your call

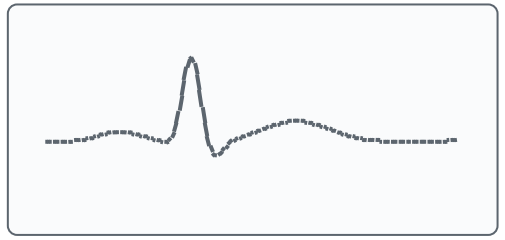
Name the family · the most likely source · one fix and how you would verify it · and what further observation would confirm the source.

Source: BME 6111 Week 5 clinic — lecturer-generated illustrative recording

3.1 Clinic — Trace T2



one recording from the patch



clean reference beat

Recording circumstance

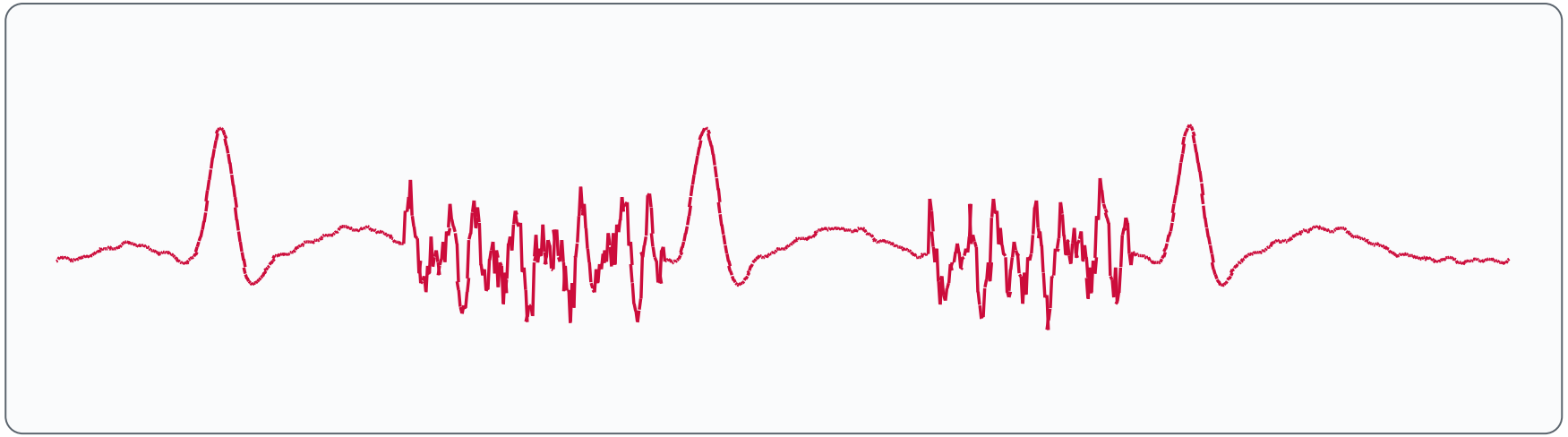
The baseline shifts during steps and steadies when the wearer stops. One patch edge has loosened.

Your call

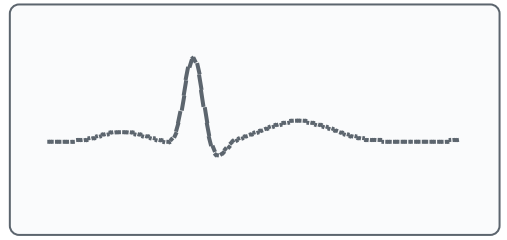
Name the family · the most likely source · one fix and how you would verify it · and what further observation would confirm the source.

Source: BME 6111 Week 5 clinic — lecturer-generated illustrative recording

3.1 Clinic — Trace T3 (extension)



one recording from the patch



clean reference beat

Recording circumstance

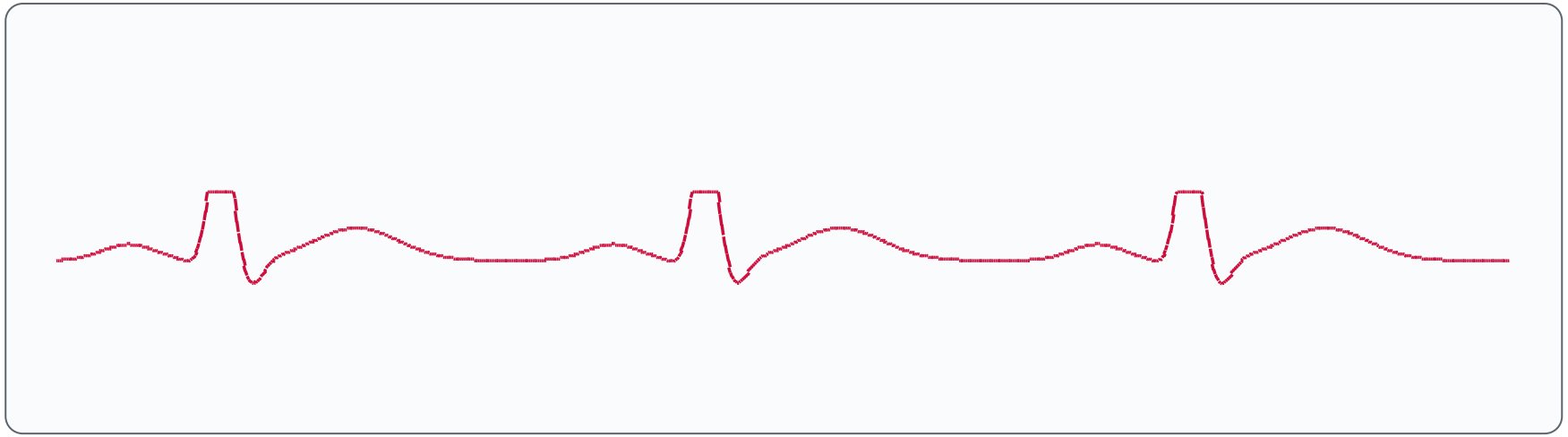
Bursts appear when the wearer tenses the chest and shoulder muscles, and subside on relaxation.

Your call

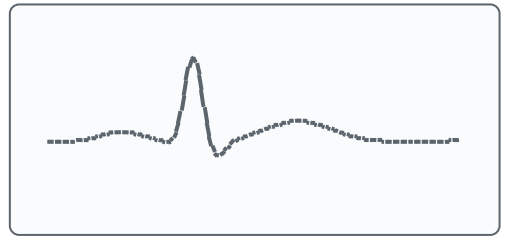
Name the family · the most likely source · one fix and how you would verify it · and what further observation would confirm the source.

Source: BME 6111 Week 5 clinic — lecturer-generated illustrative recording

3.1 Clinic — Trace T4 (extension)



one recording from the patch



clean reference beat

Recording circumstance

Peaks flatten from the start. The first stage has ±3 V supplies, a gain of 12, and an electrode offset near +300 mV.

Your call

Name the family · the most likely source · one fix and how you would verify it · and what further observation would confirm the source. First estimate what the offset alone asks of the first stage.

Source: BME 6111 Week 5 clinic — lecturer-generated illustrative recording